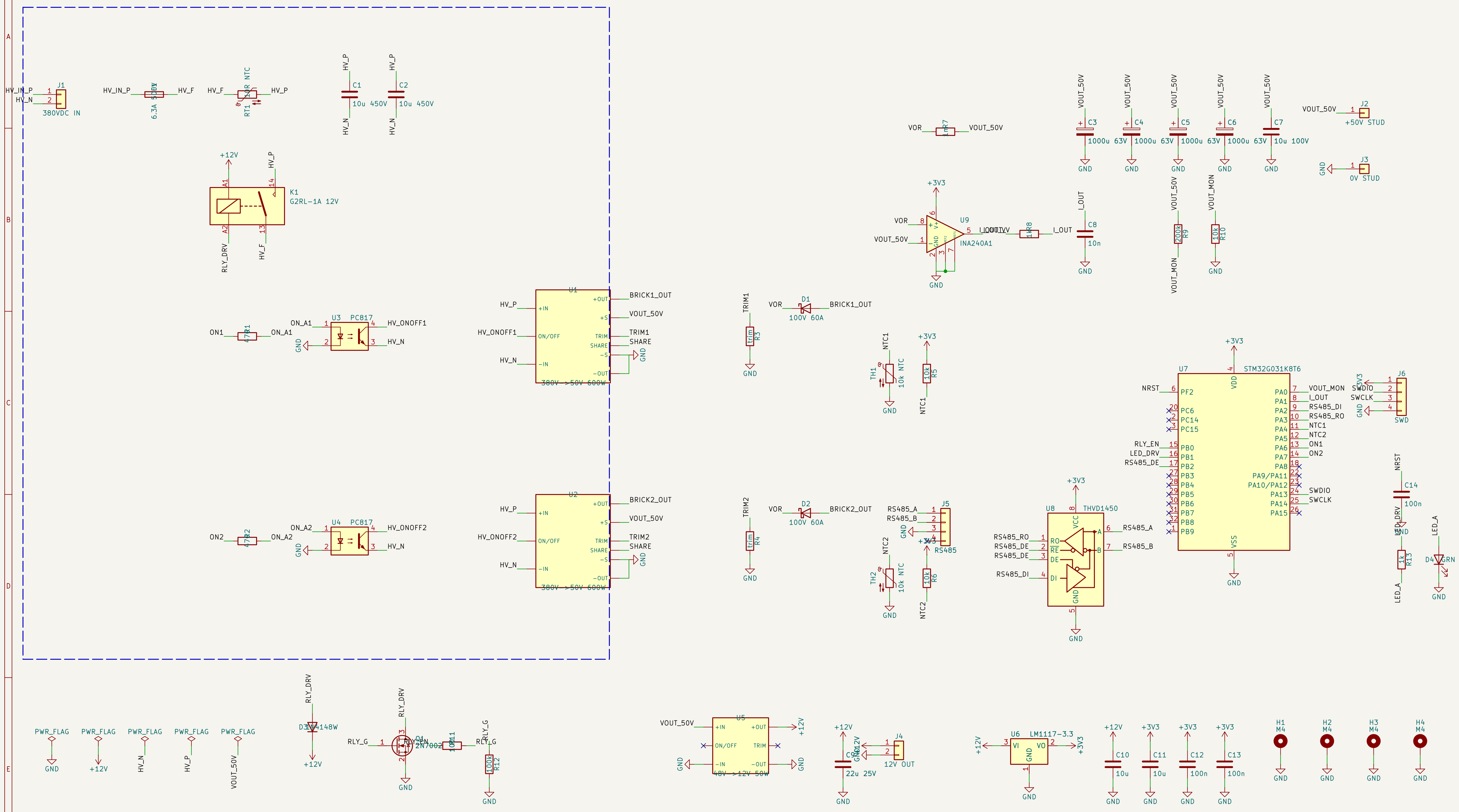
PRIMARY 380 VDC (HV_* nets: 2.5 mm clearance, isolated from secondary)

SECONDARY 50 V (pulse current flows bank <-> backplane through the studs, not this PCB)


J1
380VDC IN
HV_IN_P
HV_N
HV_IN_P
6.3A 500V
HV_F
RT1 10R NTC
HV_P
C1
10u 450V
C2
10u 450V
HV_N
+12V
K1
G2RL-1A 12V
A1
A2
14
13
RLY_DRV
ON1
R1 47R
ON_A1
U3 PC817
HV_ONOFF1
ON2
R2 47R
ON_A2
U4 PC817
HV_ONOFF2
U1
380V->50V 600W
+IN
ON/OFF
-IN
+OUT
+S
TRIM
SHARE
-S
-OUT
BRICK1_OUT
VOUT_50V
TRIM1
U2
380V->50V 600W
BRICK2_OUT
TRIM2
GND
TRIM1
R3 trim
TRIM2
R4 trim
VOR
D1
100V 60A
D2
100V 60A
R7 1m
+3V3
U9
INA240A1
I_OUT
R8 1k
C8
10n
NTC1
TH1 10k NTC
R5 10k
NTC2
TH2 10k NTC
R6 10k
J5
RS485
RS485_A
RS485_B
U8
THVD1450
RO
RE
DE
DI
VCC
RS485_RO
RS485_DE
RS485_DI
C3 1000u 63V
C4 1000u 63V
C5 1000u 63V
C6 1000u 63V
C7 10u 100V
J2
+50V STUD
J3
0V STUD
VOUT_MON
R9 200k
R10 10k
U7
STM32G031K8T6
VDD
VSS
NRST
PF2
PC6
PC14
PC15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9/PA11
PA10/PA12
PA13
PA14
PA15
RLY_EN
LED_DRV
SWDIO
SWCLK
J6
SWD
C14
100n
R13 1k
D4 GRN
LED_A
PWR_FLAG
D3 1N4148W
Q1
2N7002
RLY_G
R11 100
R12 100k
U5
48V->12V 50W
C9
22u 25V
J4
12V OUT
U6
LM1117-3.3
VI
VO
C10 10u
C11 10u
C12 100n
C13 100n
H1 M4
H2 M4
H3 M4
H4 M4

Generated by gen_schematic.py – first-pass design, not released for fabrication
SENTINEL–UHF reference design

Sheet: /
File: SENTINEL_PSU.kicad_sch

Title: SENTINEL–UHF Subarray Power Supply, 380 V –> 50 V

Size: A2
Date: 2026-10-18
Rev: A
KiCad E.D.A. 10.0.3
Id: 1/1